

Project Design Phase

Solution Architecture

Date	18 June 2026
Team ID	
Project Name	ShopEZ – MERN Stack E-Commerce Platform
Maximum Marks	4 Marks

Solution Architecture Description

ShopEZ follows a **three-tier MERN Stack Architecture** consisting of the **Presentation Layer**, **Application Layer**, and **Database Layer**.

The **Presentation Layer** is developed using **React.js, HTML, CSS, and JavaScript**, providing an interactive and responsive user interface for customers and administrators. It allows users to register, log in, browse products, search products, filter products by category, manage wishlists and shopping carts, place orders, and view order history.

The **Application Layer** is built using **Node.js and Express.js**, which handle business logic, user authentication, product management, wishlist operations, cart management, order processing, and secure communication between the frontend and backend through RESTful APIs. Authentication and authorization are implemented using **JWT (JSON Web Tokens)** to ensure secure access to protected resources.

The **Database Layer** uses **MongoDB Atlas** to store and manage user accounts, product information, wishlist items, cart details, and order records. MongoDB provides flexible and scalable data storage for the e-commerce platform.

The architecture enables secure and efficient communication between users, administrators, and the database through RESTful APIs. This design improves scalability, maintainability, reliability, and performance while providing a seamless online shopping experience for customers and efficient product management capabilities for administrators.

Solution Architecture Diagram:

